

AMACE Version 2.0
Optimized Patent Specification

Autonomous Multi-Stratum Anomaly Correlation Engine.
A domain-agnostic, zero-configuration framework for
counterfactually-validated causal discovery.

INFERENCE
STRATUM

S₂

REPRESENTATION
STRATUM

S₁

OBSERVATION
STRATUM

S₀

CROSS-STRATUM
ANOMALY EVENT

CAUSAL INFERENCE VALIDATION:
Counterfactual Impact Analysis & Root
Cause Identification

LATENT ANOMALY CORRELATION:
Feature Shift & Representation Collapse

ANOMALY DETECTION:
Deviation from Baseline, $t=t_{anomaly}$

CROSS-STRATUM ANOMALY

COLOUR LEGEND:

YELLOW = OBSERVATION STRATUM (S₀)

PURPLE = REPRESENTATION STRATUM (S₁)

ORANGE = INFERENCE STRATUM (S₂)

CHARCOAL = STRUCTURAL/ANOMALY LINES

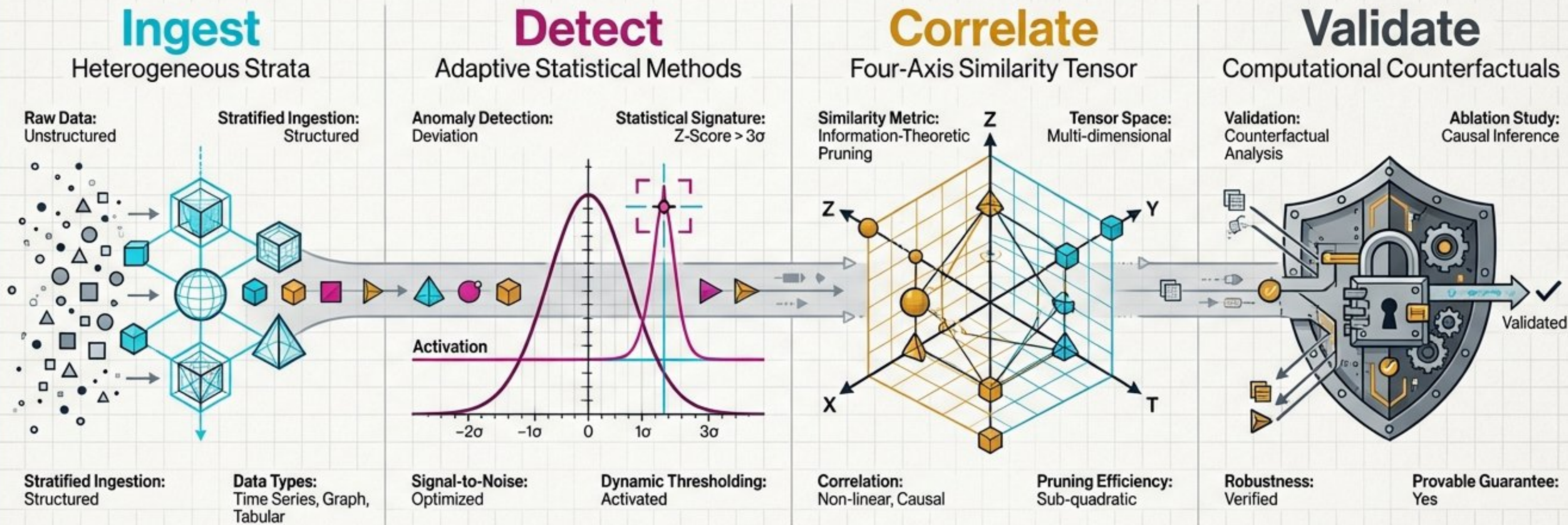
SLATE GREY = TENSOR GRID/TEST

PROJECT: AMACE v3.0 // DRAWING NO: 801

DATE: OCT 26, 2023 // ENGINEER: J. DOE, PhD

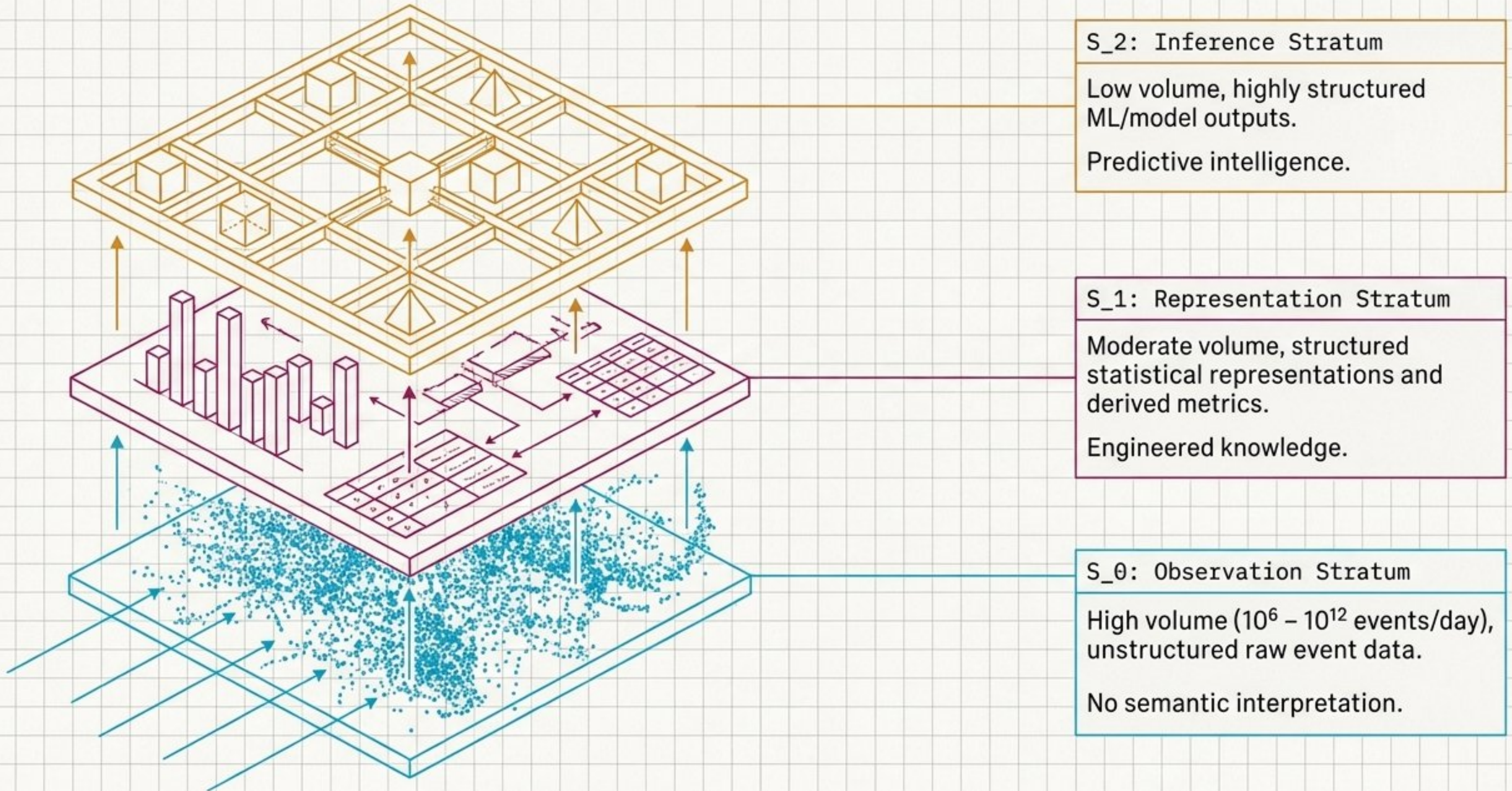
STATUS: CLASSIFIED // PATENT PENDING

Autonomous multi-stratum causal discovery without human configuration



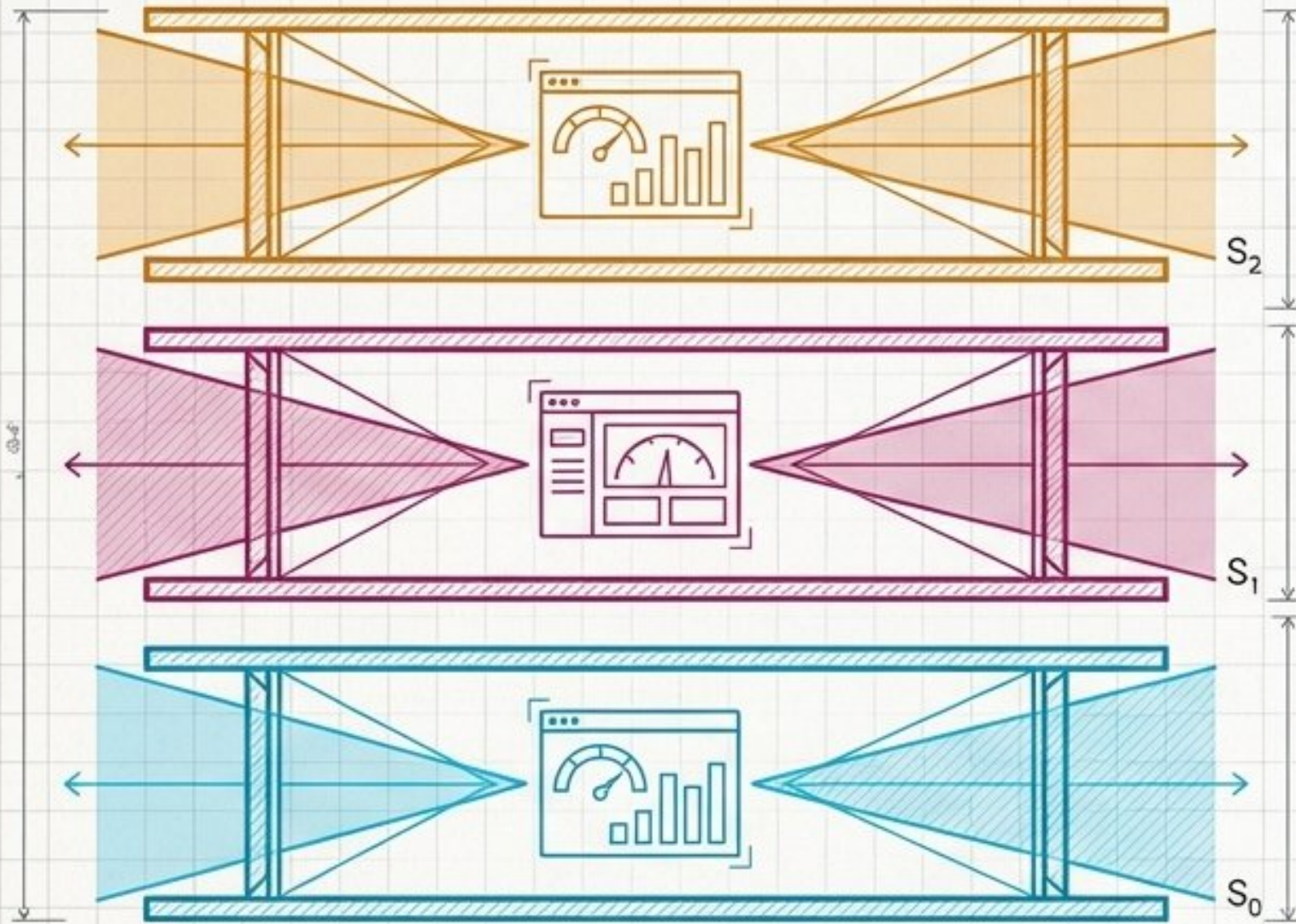
AMACE is the first system to ingest heterogeneous data, detect stratum-specific anomalies, correlate them via information-theoretic pruning, and validate causality through computational ablation—provably within sub-quadratic computational complexity.

Modern computing systems operate across three isolated abstraction strata

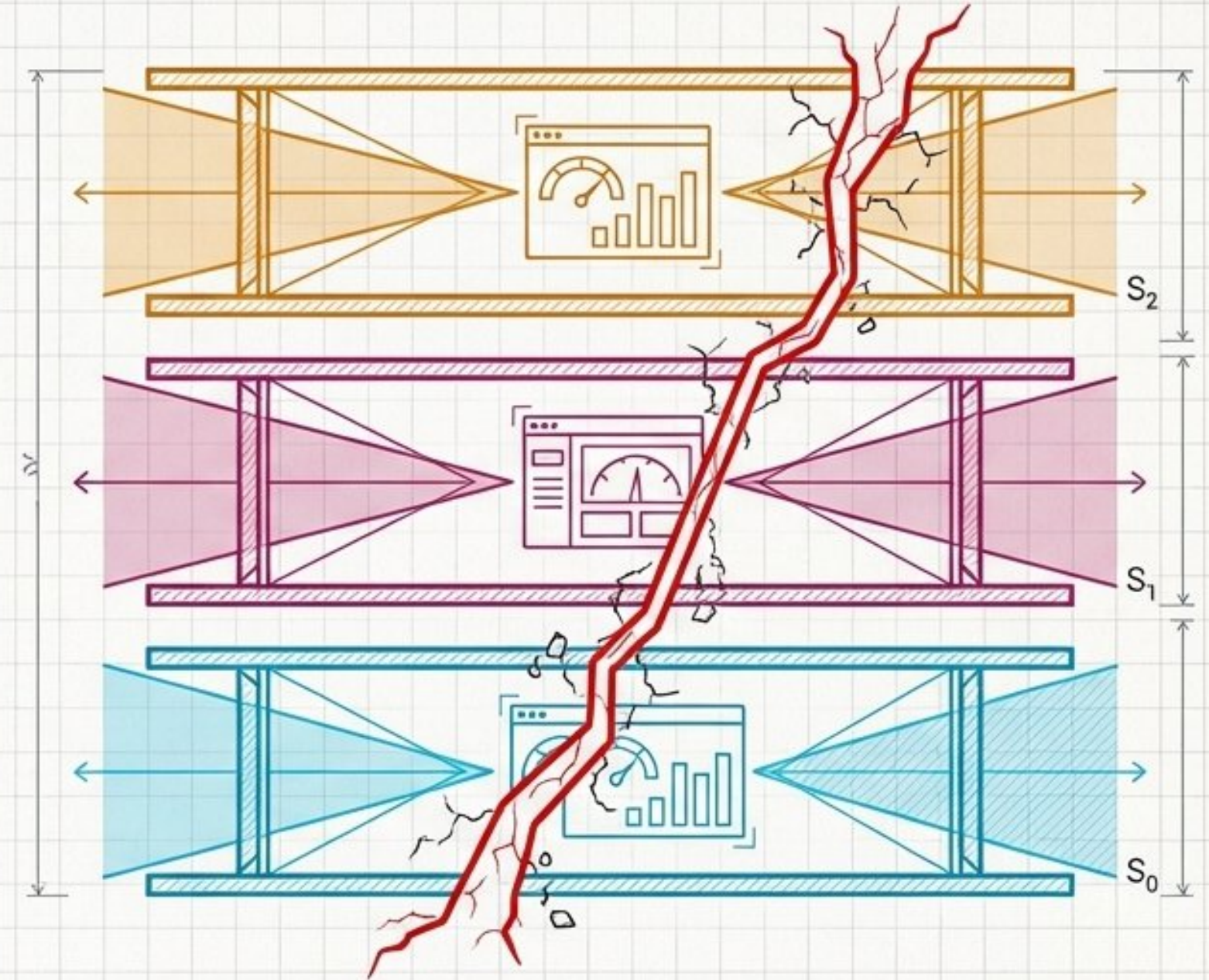


Existing monitoring tools are horizontally blind to vertical causal cascades

The Horizontal View



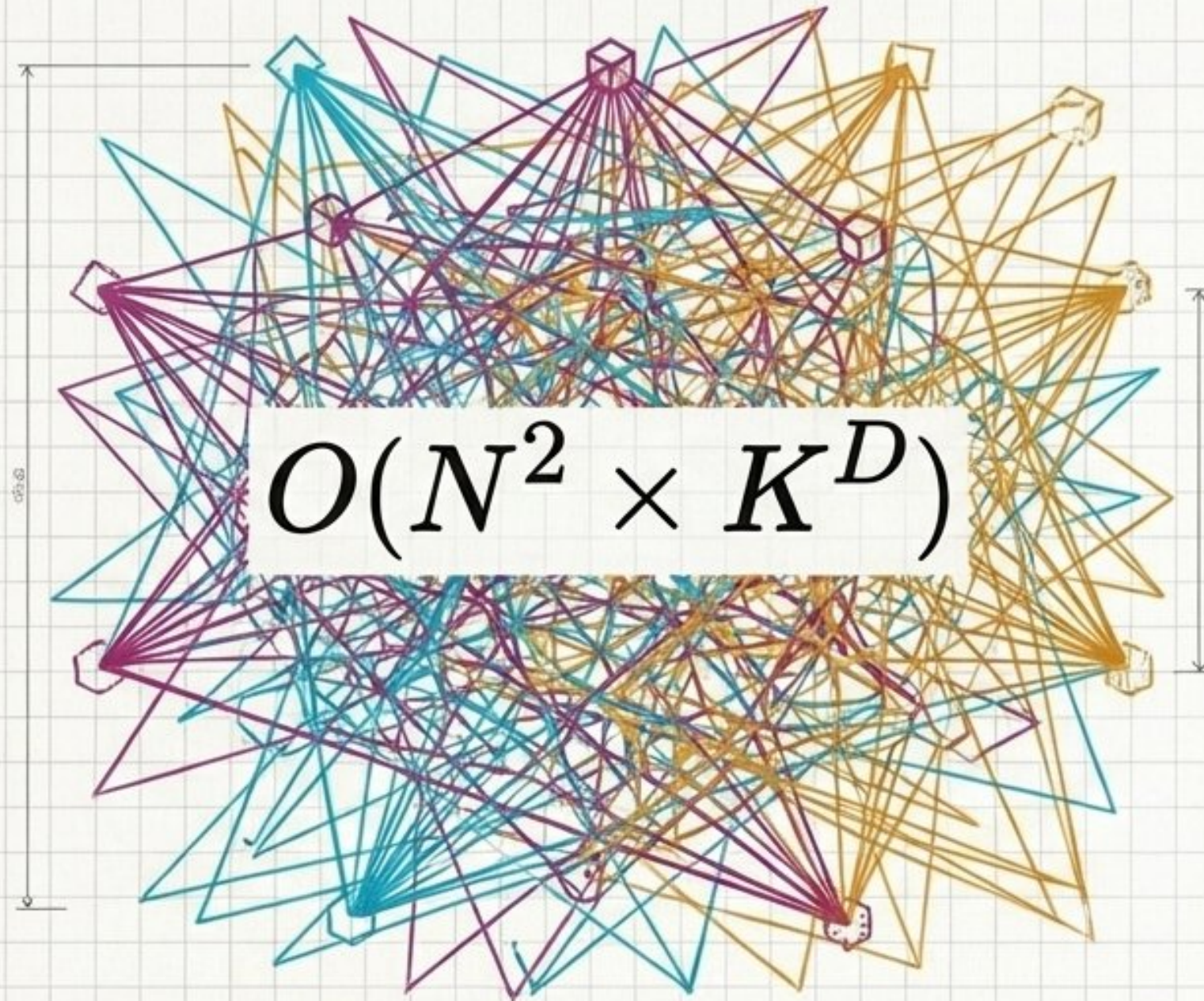
The Vertical Cascade



When a malfunction causes simultaneous deviations across S_0 , S_1 , and S_2 , no existing tool can determine they share a causal origin. Each tool sees only its own stratum's symptoms.

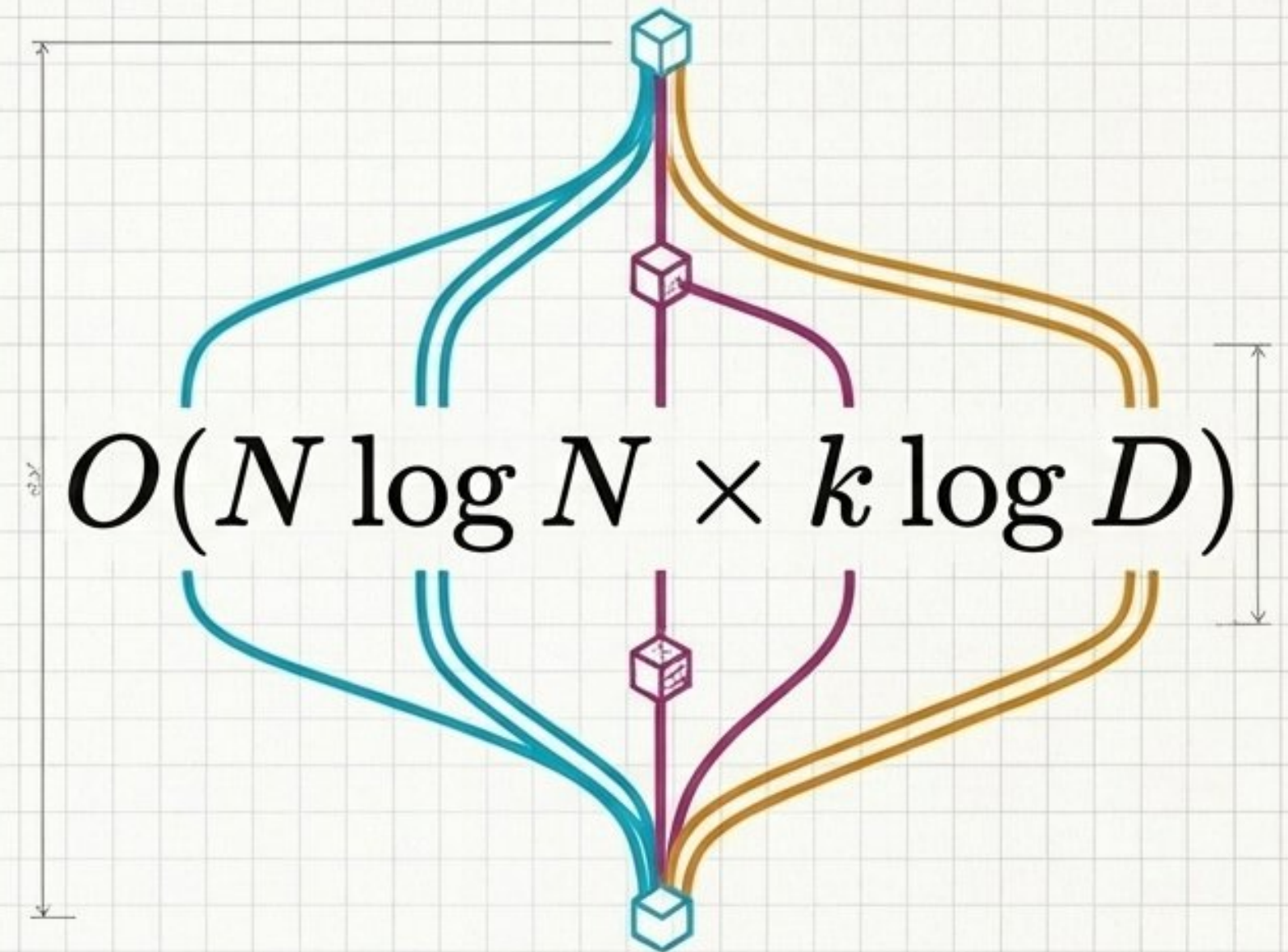
Exhaustive cross-stratum correlation is mathematically intractable

The Problem



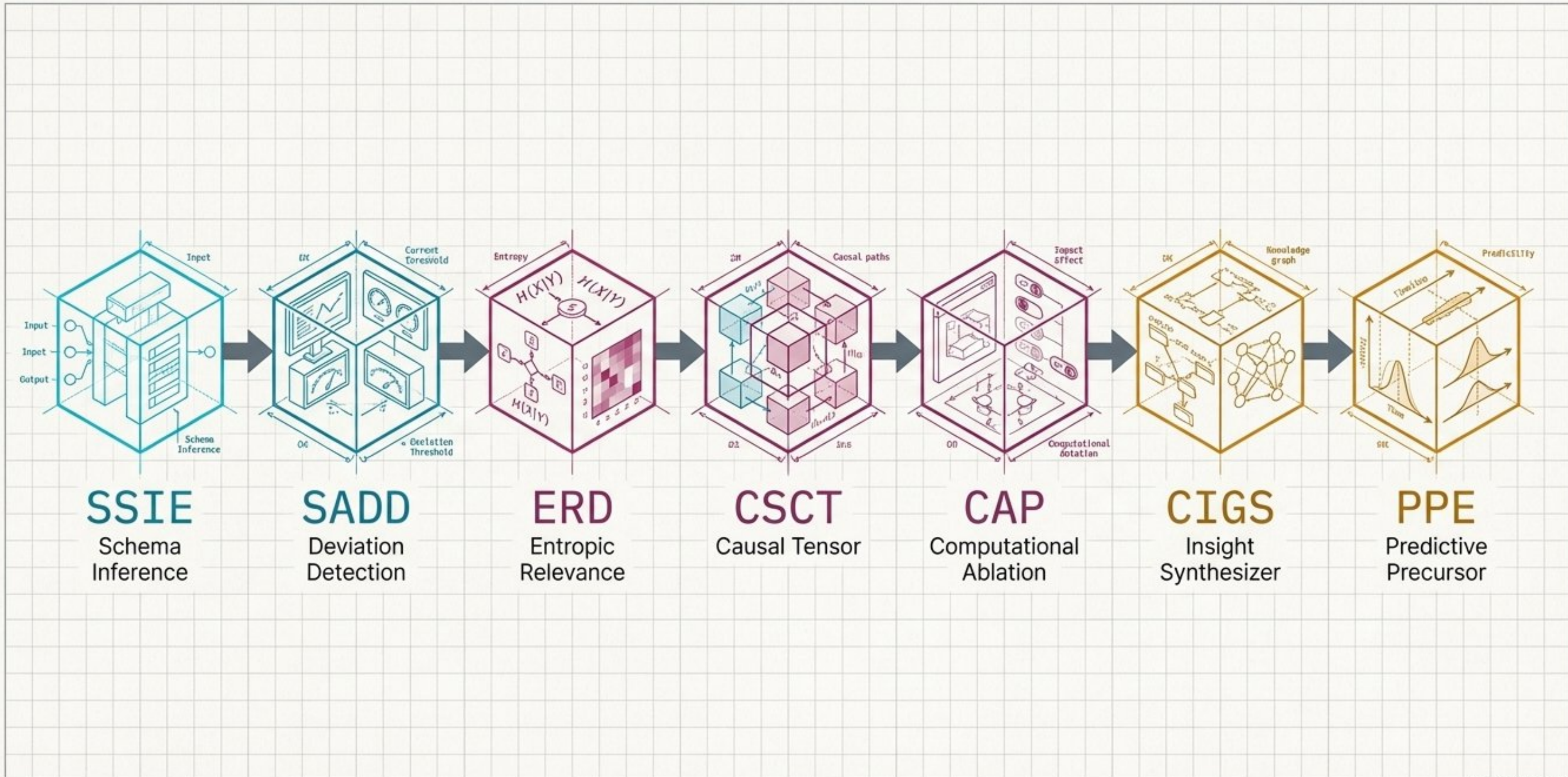
For typical production systems, this exceeds 10^{18} combinations. Computationally infeasible.

The AMACE Solution

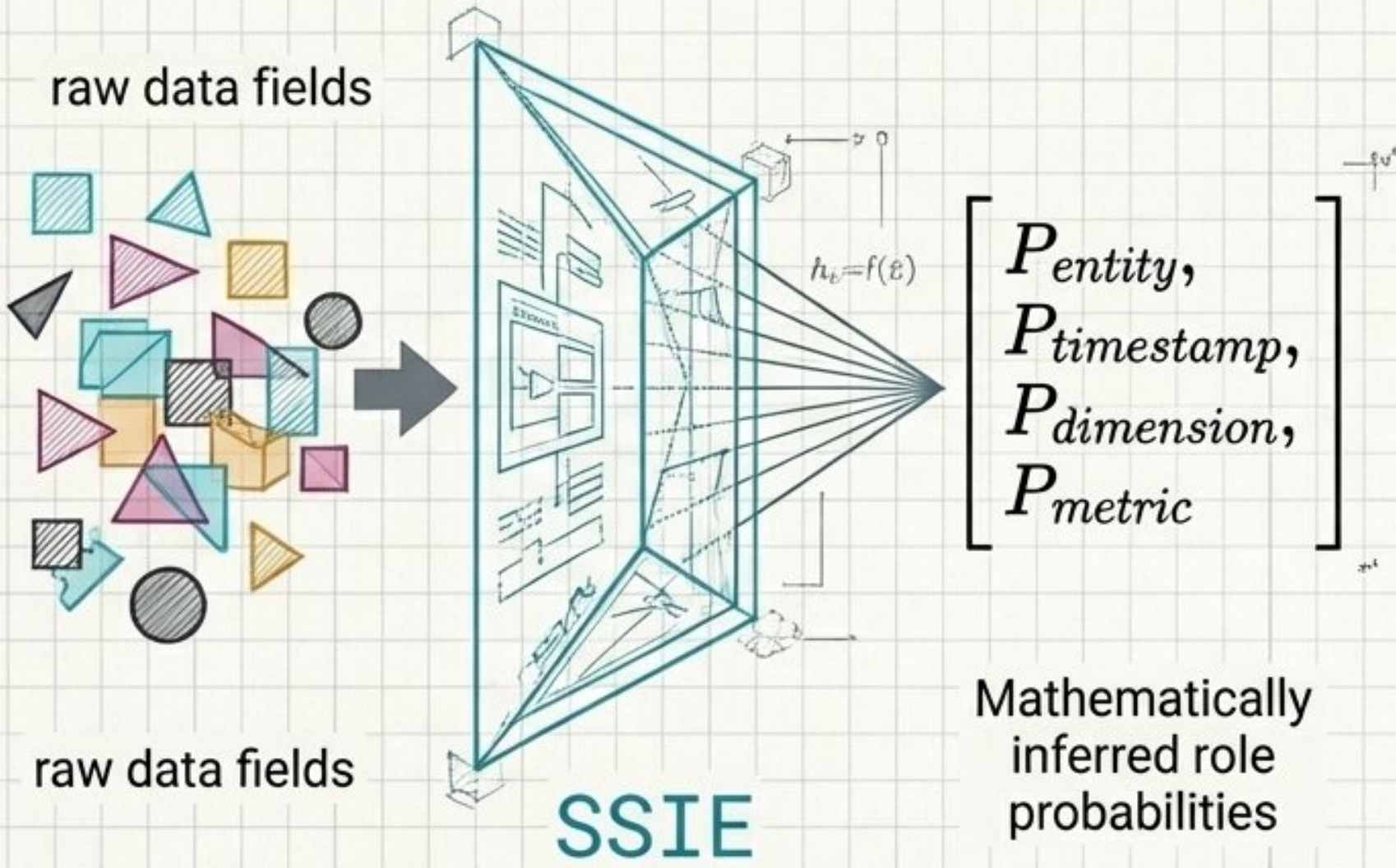


Exponential reduction achieved by isolating only informationally relevant dimensional slices ($k \ll K^D$).

A mathematically bounded 7-module pipeline for causal discovery



SSIE mathematically fingerprints data roles without human configuration



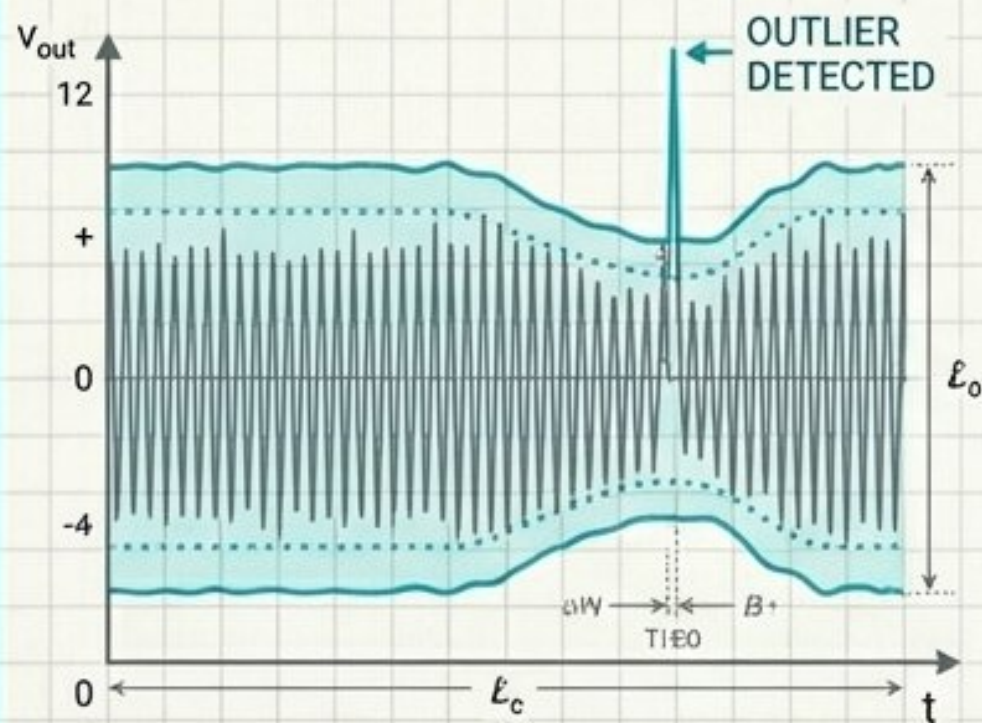
Data Role	Statistical Fingerprint
Entity	High Shannon entropy, high cardinality.
Timestamp	Low inter-arrival delta entropy, monotonic.
Dimension	Temporal stability, repeated values.
Metric	Continuous distribution, finite support.

SADD adapts statistical detection to the distinct nature of each stratum



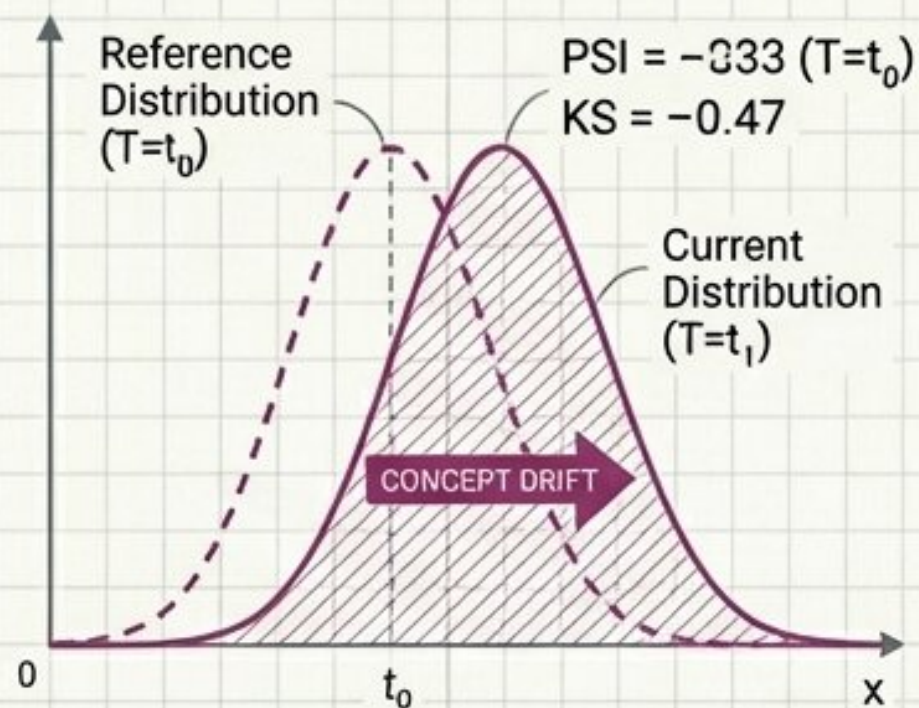
S_0 - Observation

Modified Z-score with Median Absolute Deviation (MAD) for outlier robustness. Auto-tuned to detected periodicity.



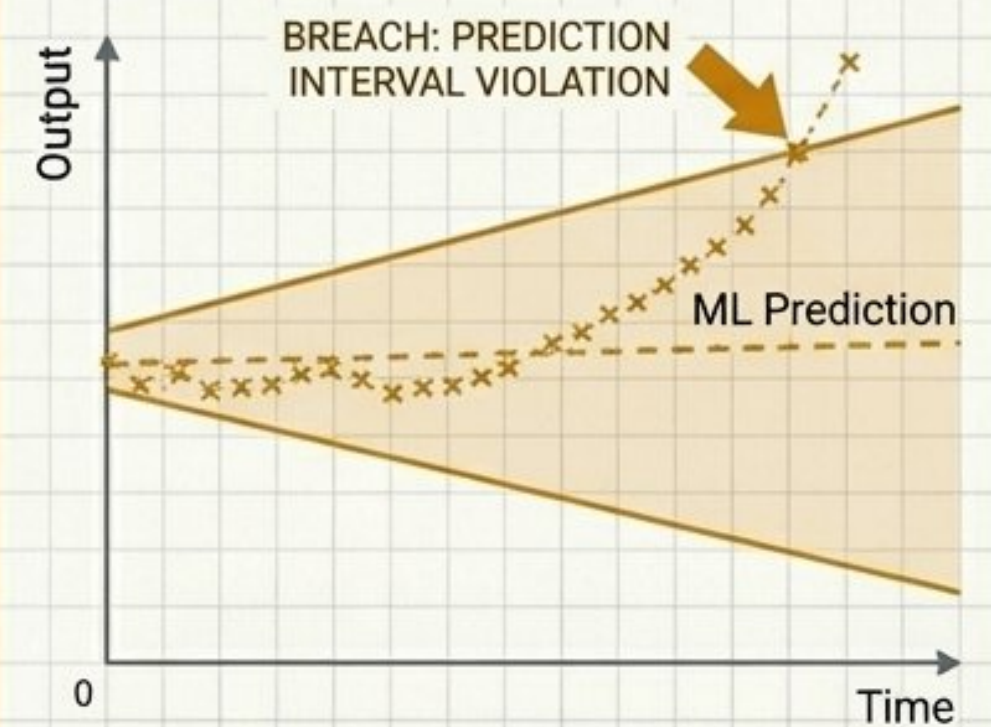
S_1 - Representation

Population Stability Index (PSI) with two-sample Kolmogorov-Smirnov confirmation. Tracks concept drift.

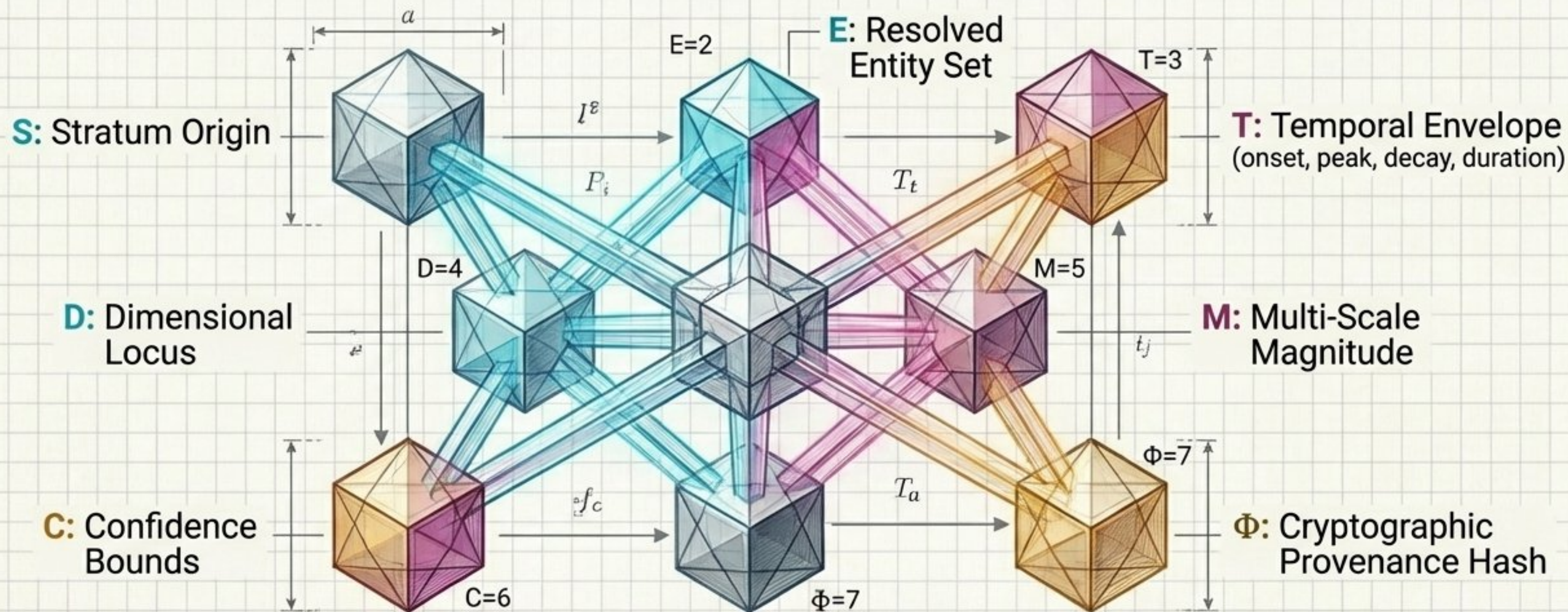


S_2 - Inference

Prediction Interval Monitoring. Tracks whether ML outputs breach dynamically calibrated confidence bounds.

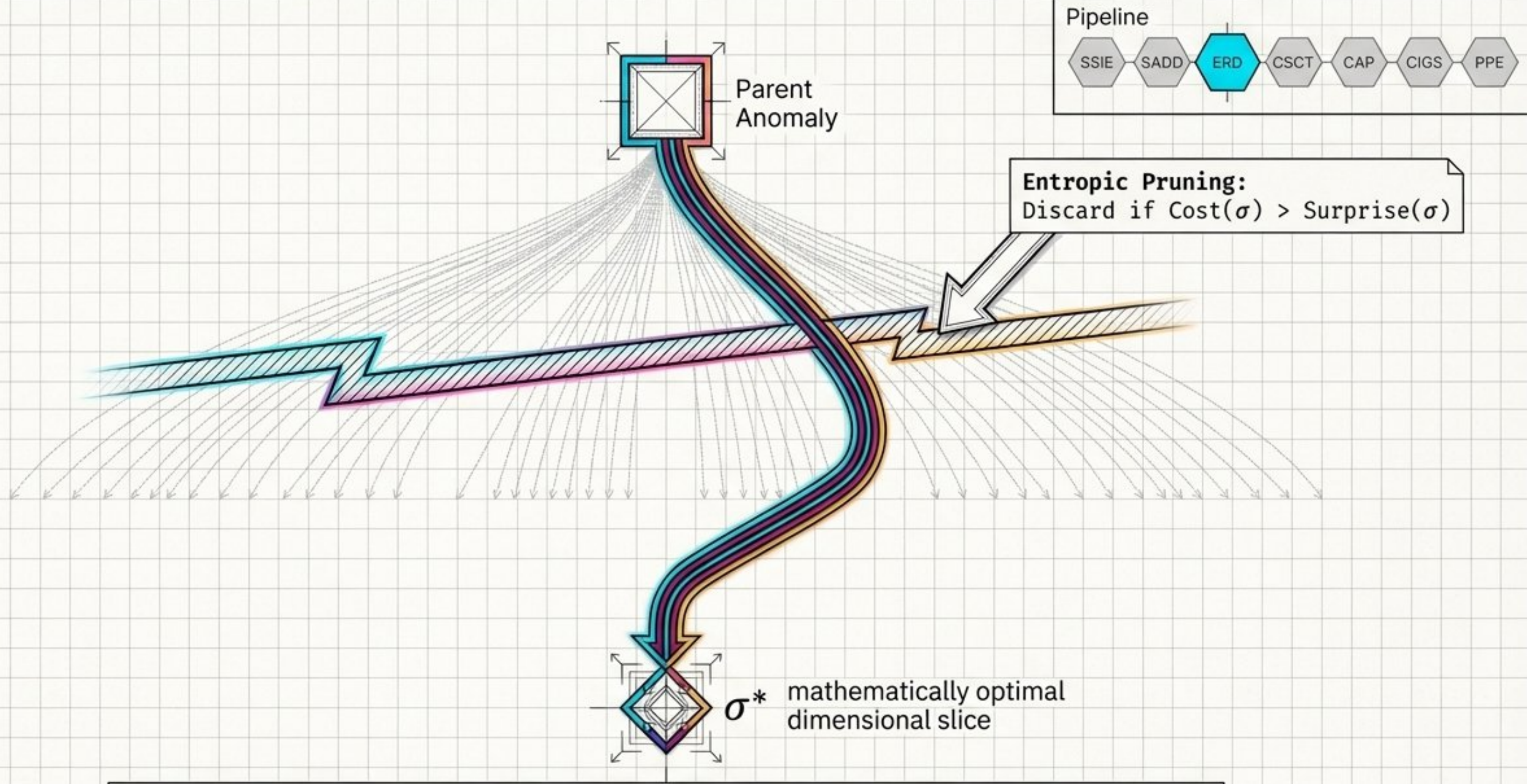


The Deviation Signature Tensor: The universal DNA of an anomaly



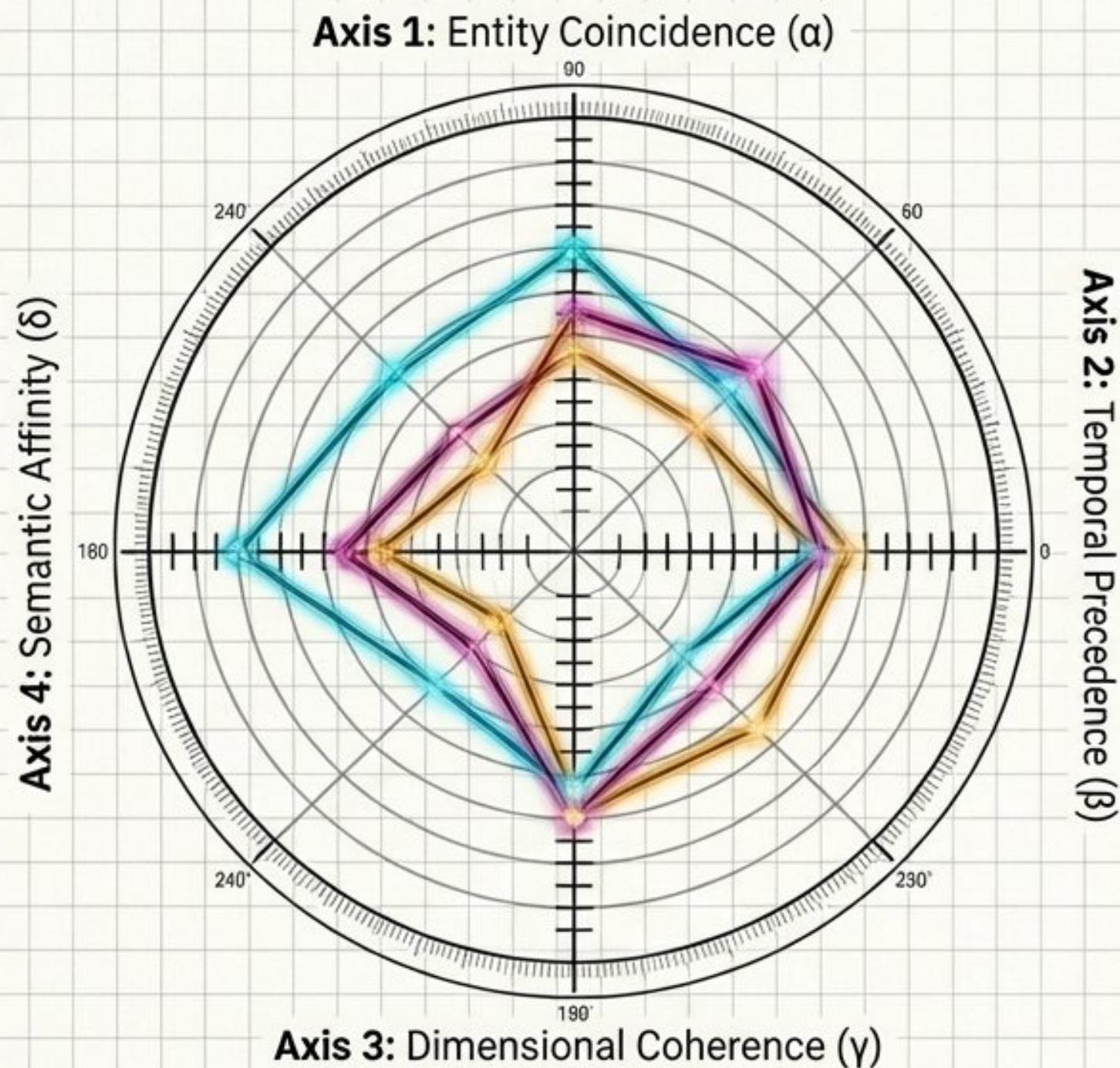
No prior art defines a universal anomaly representation encoding **stratum origin**, **multi-scale magnitude**, and **cryptographic provenance** in a single algebraic structure.

ERD uses entropic beam search to prune 99.9% of the dimensional space



Achieves $O(w \times D \times K_{max})$ time complexity.
For typical systems, this drops evaluations from 10^{16} to **40,000**.

CSCT establishes causal links through a four-axis similarity tensor



Pipeline



Axis 1: Entity Coincidence (α)

Surprise-adjusted Jaccard overlap.

Axis 2: Temporal Precedence (β)

Gaussian kernel learned propagation delay.

Axis 3: Dimensional Coherence (γ)

Hierarchical parent-child matching.

Axis 4: Semantic Affinity (δ)

Contrastive learning metric embeddings.

Axes are combined using spectral normalization. Weights dynamically adapt based on signal characteristics (e.g., sudden onset vs. gradual drift).

CAP mathematically proves causation through computational counterfactuals

Pipeline



Test 1: Exclusivity Ablation

Remove DST A. Does Anomaly *B* disappear?
(Checks if A is necessary)



Test 2: Sufficiency Ablation

Use only DST A. Can it accurately predict *B*?
(Checks if A is sufficient)



Test 3: Minimality Ablation

Iteratively remove elements from a multi-DST chain.
What is the smallest required set?
(Prevents over-attribution)

Each ablation runs with 1,000 bootstrapped iterations.
Bounding Type I error rate at $\alpha = 0.05$.

Translating validated causal math into DAGs and predictive warnings

PIPELINE MINI-MAP

SSIE

SADD

ERD

CSCT

CAP

CIGS

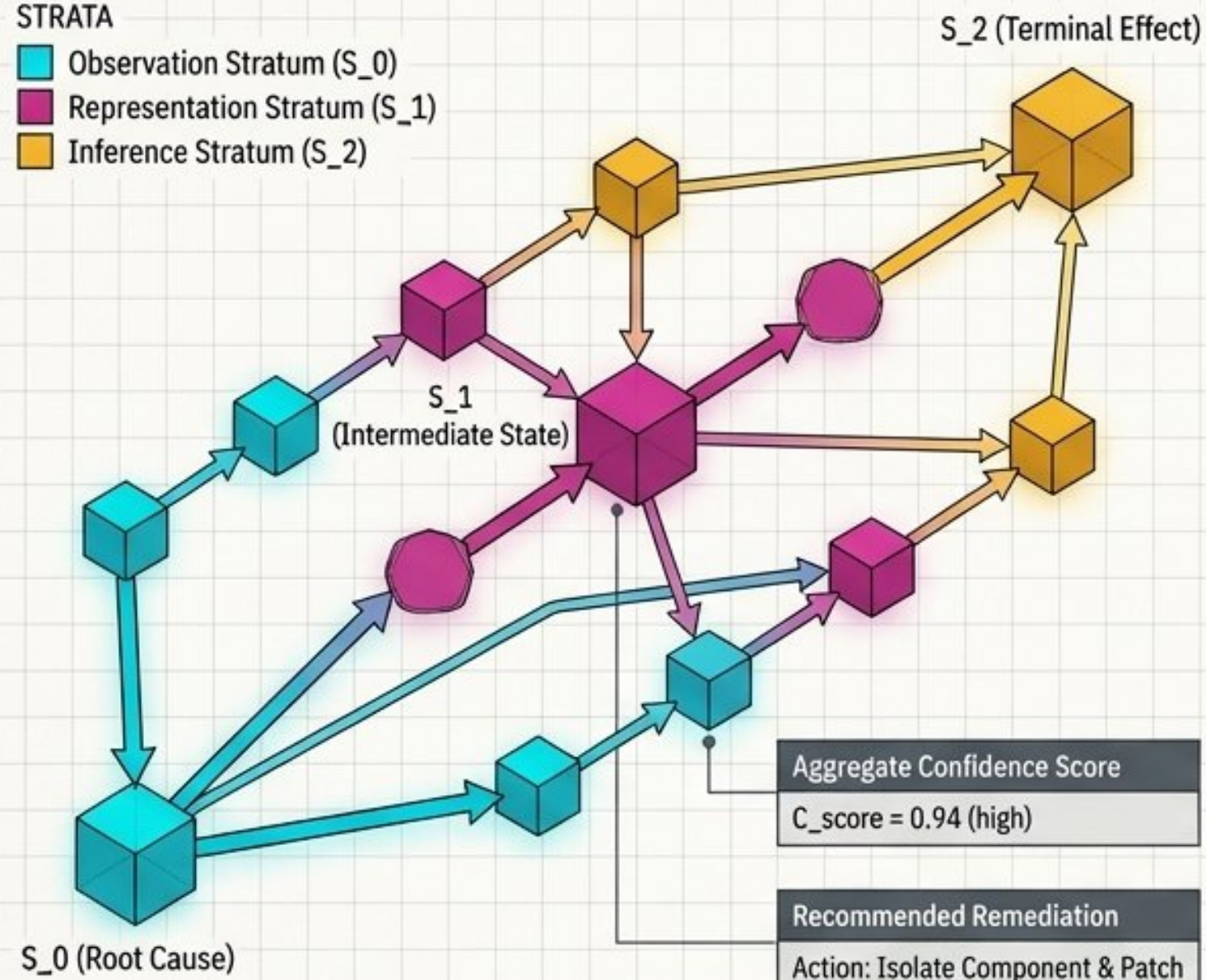
PPE

CIGS - Insight Graph Synthesizer

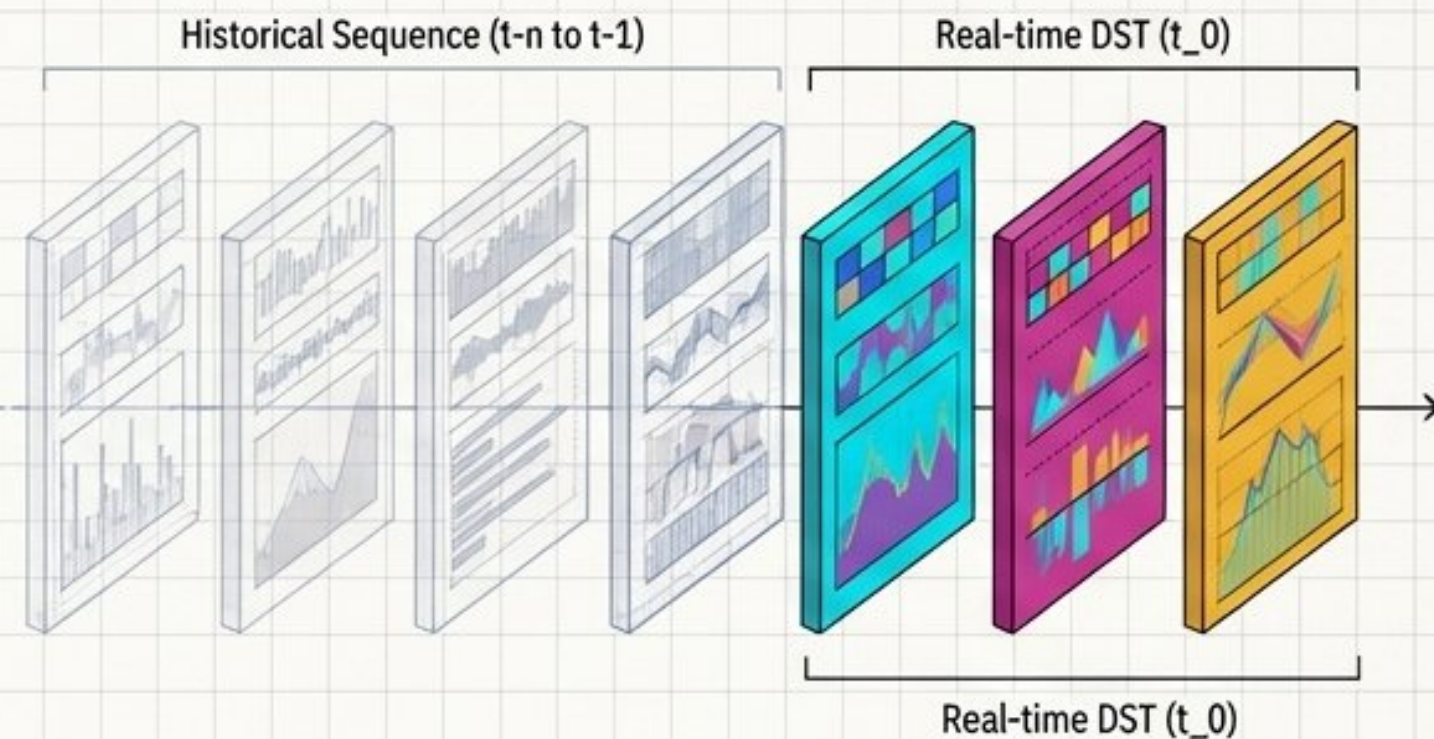
STRATA

 Observation Stratum (S₀)

 Representation Stratum (S₁)

 Inference Stratum (S₂)


PPE - Predictive Precursor Engine


 $P(\text{incident} \mid \text{prefix})$
 $P(\text{incident} \mid \text{prefix}) = 0.89 > P_{\text{thresh}}$

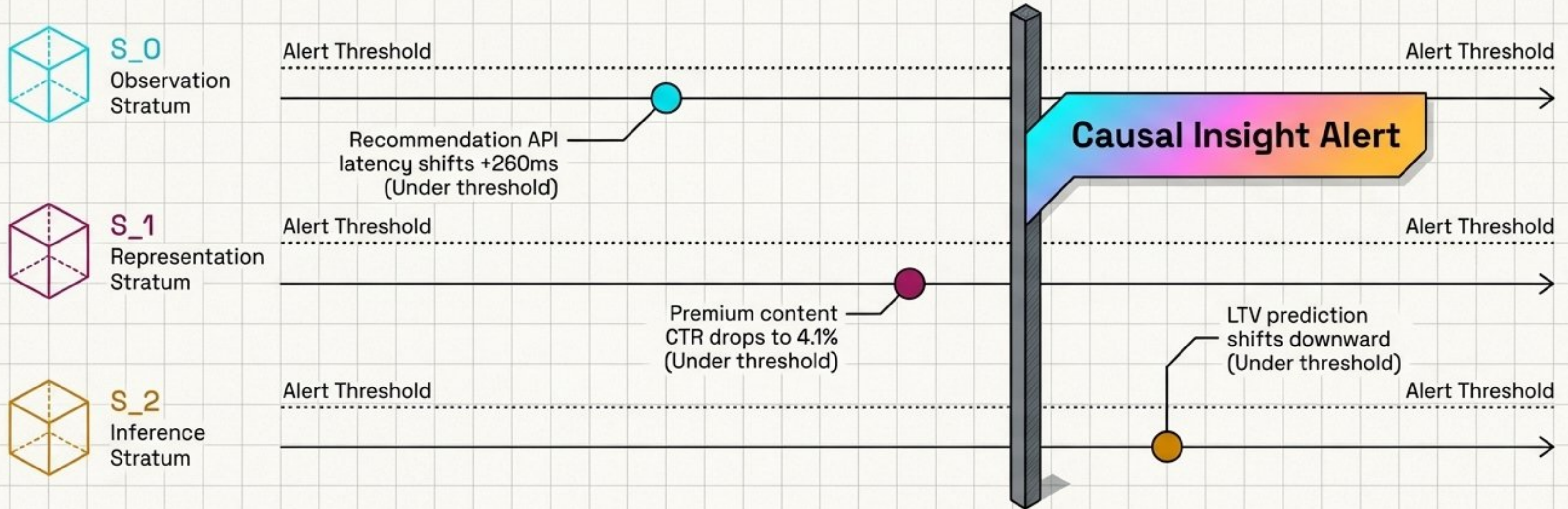
Incident Probability Accumulation

The three-stratum model maps perfectly across any heterogeneous domain

Universal Applicability Matrix					
Domain	S_0 (Observation) 		S_1 (Representation) 		S_2 (Inference) 
Telecom	Raw packet captures	→	Jitter percentiles	→	Congestion predictions
Mobile App	API response logs	→	CTR drops	→	Churn prediction shifts
Cloud Infra	Kubernetes events	→	Pod CPU usage	→	Autoscaling decisions
Finance	FIX protocol ticks	→	Volatility surfaces	→	VaR models
Healthcare	Vital sign captures	→	Adherence indices	→	Readmission risks

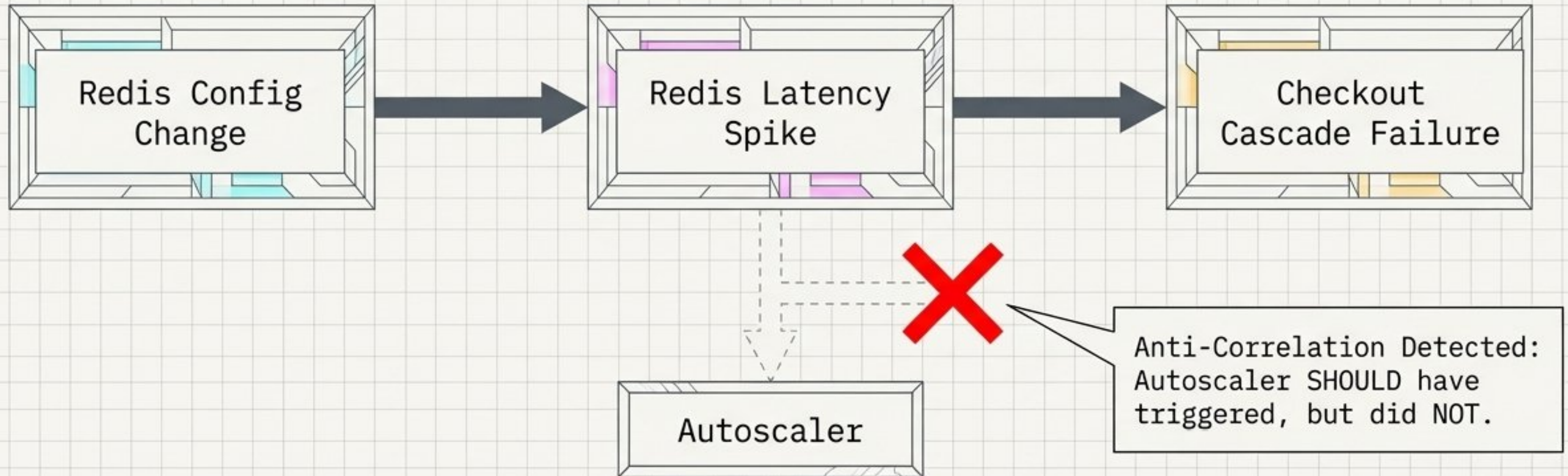
Correlating sub-threshold anomalies reveals a \$2.4M invisible failure

Mobile App Deep Dive



Because no single metric breached its individual threshold, the \$2.4M leak was completely invisible to existing observability stacks.

Discovering latent bugs by computationally tracking what did not happen



AMACE not only identifies the root cause of the active incident but detects missing expected correlations to surface silent infrastructure misconfigurations.

Version 2.0 re-engineers vocabulary for strict patent defensibility

Before (v1) vs. After (v2.0)



Generic v1 Concept

Bounded v2.0 Mathematical Construct

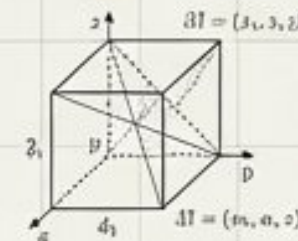
Generic Layers

Precise Strata and Cross-Stratum Causal Tensors



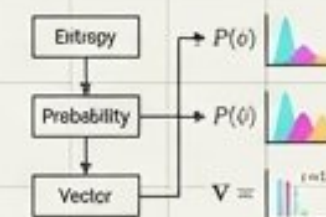
JSON Ad-hoc formats

Formally defined 7-dimension Deviation Signature Tensor (DST)



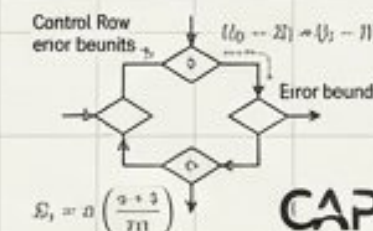
Auto-discover

Entropy-based field classification probability vectors



Counterfactuals

Computational Ablation Protocol (CAP) with Type I/II error bounds



Generic terms are easily mapped to prior art. Version 2.0 utilizes bounded mathematical constructs that cannot be abstracted.

Mathematical bounds and unexpected results
shield against statutory challenges

Surviving 101 (Alice / Abstract Idea)

- DST Data Structure (provenance hashing)
- Entropic Relevance Decomposition
- Exponential reduction of computational overhead

Defense: Concrete technical improvement to computer functioning.

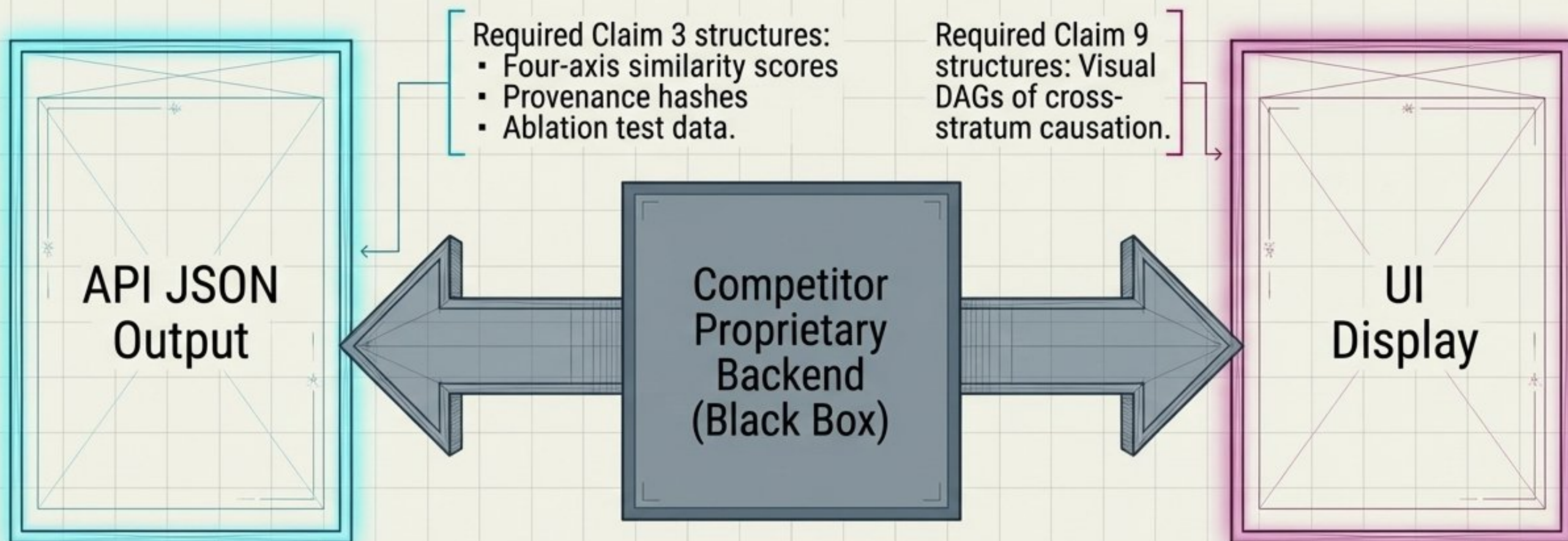
Surviving 103 (Obviousness)

- 99.97% Search Space Pruning without precision loss
- Sub-threshold cascade detection
- Anti-Correlation mechanics

Defense: Teaching-away evidence from the \$50B monitoring industry.

Infringement can be proven strictly through externally observable outputs

"Outside-In" Enforceability Map



Every internal AMACE module claims a machine-specific, externally observable output.
Infringement detection requires zero access to source code.

HIGHLY CLASSIFIED TECHNICAL DOSSIER

ARCHITECTURAL BLUEPRINT

Claim 1: Core Method

The foundational computational sequence from SSIE ingestion through CAP ablation.

Claim 2: System Claim

The specific hardware/memory execution of the mathematical bounds.

Claim 3: Observable Output

The structured JSON artifact guaranteeing enforceable API protection.

Claim 4: Predictive Warning

The temporal sequence matching that captures pre-Incident precursor states.

STATE

STATE

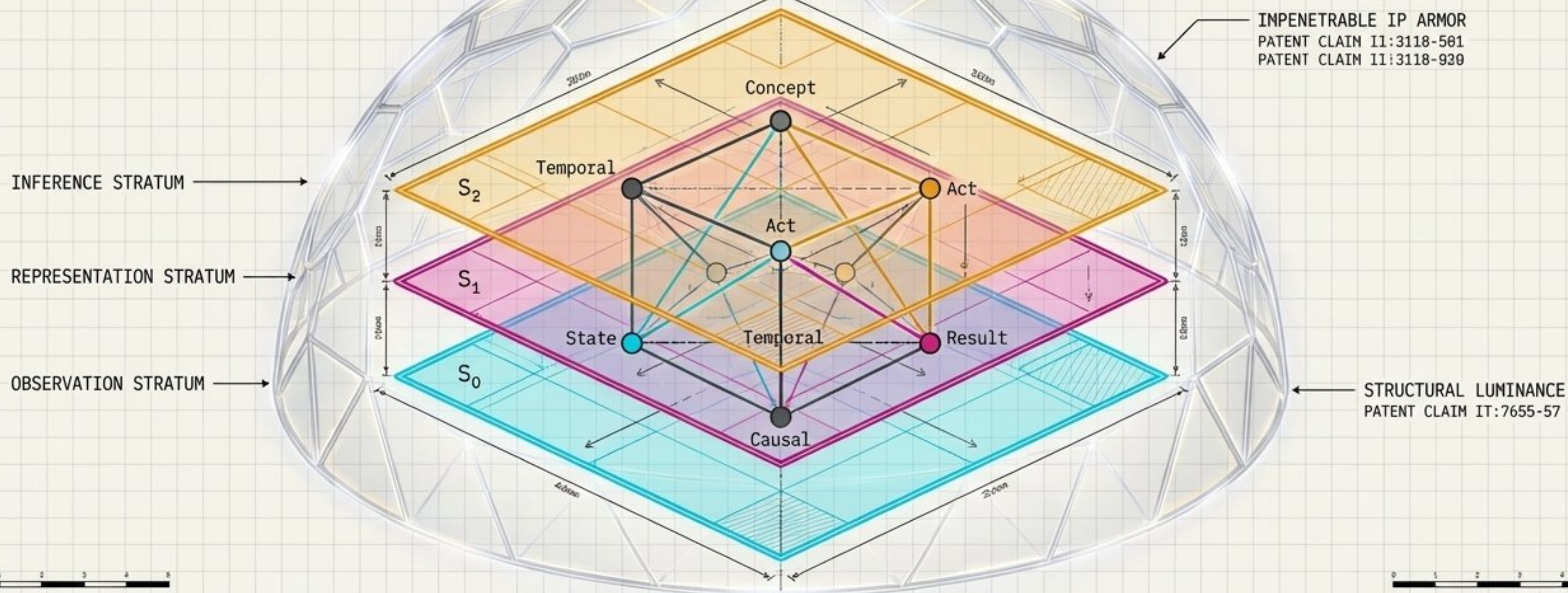
STATE

STATE

Bridging the gap between deep mathematical theory and commercial IP

HIGHLY CLASSIFIED TECHNICAL DOSSIER

ARCHITECTURAL BLUEPRINT



AMACE reduces the intractable 10^{18} complexity of cross-stratum correlation into an actionable $O(N \log N)$ pipeline—safeguarded by externally enforceable patent claims that fundamentally disrupt the observability market.